RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 6 (Delta)
Topic(s): Vectors
(Vectors II and III (Solutions))

Source of Question: JPJC Prelim 9758/2024/01/Q6 ii and iii(modified)

1 Referred to the origin O, points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively such that angle $AOB = 30°$ and $|\mathbf{a}| = 4|\mathbf{b}|$.

Point D is the mid-point of OA and point E lies on OB such that $OE : EB = 3 : 2$.

(i) Find the position vectors $\overrightarrow{OD}$ and $\overrightarrow{OE}$, giving your answers in terms of $\mathbf{a}$ and $\mathbf{b}$. [2]

(ii) Show that the vector equation of the line BD can be written as $\mathbf{r} = \lambda\mathbf{a} + (1-2\lambda)\mathbf{b}$, where λ is a parameter. By finding the vector equation of the line AE in a similar form in terms of a parameter μ, find, in terms of $\mathbf{a}$ and $\mathbf{b}$, the position vector of the point F where the lines BD and AE meet. [4]

Solution:

(i)

$$\overrightarrow{OD} = \frac{1}{2}\underset{\sim}{a}$$

$$\overrightarrow{OE} = \frac{3}{5}\underset{\sim}{b}$$

(ii)

$$\begin{aligned} l_{BD} : \underset{\sim}{r} &= \underset{\sim}{b} + s\overrightarrow{BD} \\ &= \underset{\sim}{b} + s(\underset{\sim}{d} - \underset{\sim}{b}) \\ &= \underset{\sim}{b} + s\left(\frac{1}{2}a - \underset{\sim}{b}\right) \\ &= \underset{\sim}{b} + \lambda(\underset{\sim}{a} - 2\underset{\sim}{b}), \quad \text{where } \lambda = \frac{s}{2} \\ &= \lambda\underset{\sim}{a} + (1-2\lambda)\underset{\sim}{b} \quad \text{(shown)} \end{aligned}$$

$$\begin{aligned} l_{AE} : \underset{\sim}{r} &= \underset{\sim}{a} + t\overrightarrow{AE} \\ &= \underset{\sim}{a} + t(\underset{\sim}{e} - \underset{\sim}{a}) \\ &= \underset{\sim}{a} + t\left(\frac{3}{5}\underset{\sim}{b} - \underset{\sim}{a}\right) \\ &= \underset{\sim}{a} + \mu(3\underset{\sim}{b} - 5\underset{\sim}{a}), \quad \text{where } \mu = \frac{t}{5} \\ &= (1-5\mu)\underset{\sim}{a} + 3\mu\underset{\sim}{b} \end{aligned}$$

Since lines *BD* and *AE* meet,

$$\lambda\underset{\sim}{a}+(1-2\lambda)\underset{\sim}{b}=(1-5\mu)\underset{\sim}{a}+3\mu\underset{\sim}{b}$$

Since $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are non-zero and non-parallel vectors,

$\lambda=1-5\mu \Rightarrow \lambda+5\mu=1$ --- (1)

$1-2\lambda=3\mu \Rightarrow 2\lambda+3\mu=1$ --- (2)

Using GC, $\lambda=\frac{2}{7}, \mu=\frac{1}{7}$.

$$\overrightarrow{OF}=\frac{2}{7}\underset{\sim}{a}+\left[1-2\left(\frac{2}{7}\right)\right]\underset{\sim}{b}$$

$$=\frac{2}{7}\underset{\sim}{a}+\frac{3}{7}\underset{\sim}{b}$$

Source of Question: YIJC Prelim 9758/2025/P1/Q9

2 Planes p and q are perpendicular. Plane p has equation $\mathbf{r}\cdot(\mathbf{i}+2\mathbf{j}-\mathbf{k})=6$. Plane q contains the line l with equation $\mathbf{r}=\mathbf{j}-12\mathbf{k}+\lambda(4\mathbf{i}+\mathbf{j}+10\mathbf{k})$ where λ is a parameter.

(a) Find a cartesian equation of q. [2]

(b) Find a vector equation of the line m, where p and q meet. [2]

Point A has coordinates $(0,1,-12)$. l passes through p at point B. Point C is the foot of the perpendicular of A to p.

(c) Find the position vector of B. [2]

(d) Find the position vector of C. [2]

(e) Find the exact area of triangle ABC. [2]

Solution:

(a)

Since planes p and q are perpendicular, to find normal of q,

$$\begin{pmatrix}1\\2\\-1\end{pmatrix}\times\begin{pmatrix}4\\1\\10\end{pmatrix}=\begin{pmatrix}20-(-1)\\-(10-(-4))\\1-8\end{pmatrix}=\begin{pmatrix}21\\-14\\-7\end{pmatrix}=7\begin{pmatrix}3\\-2\\-1\end{pmatrix}$$

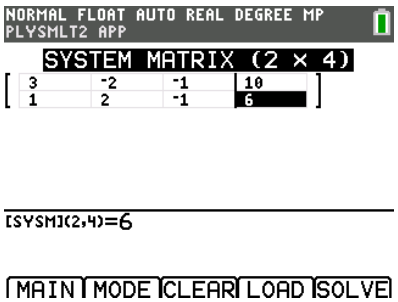
$$\mathbf{r}\bullet\begin{pmatrix}3\\-2\\-1\end{pmatrix}=\begin{pmatrix}0\\1\\-12\end{pmatrix}\bullet\begin{pmatrix}3\\-2\\-1\end{pmatrix}=10$$

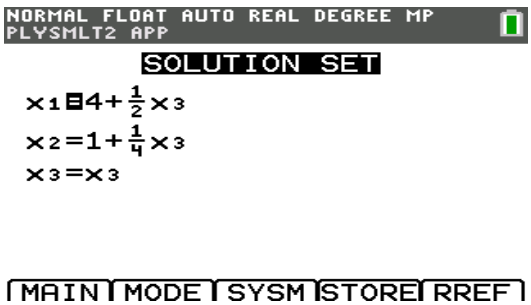
$$3x-2y-z=10$$

(b)

A cartesian equation of q is $3x-2y-z=10$

A cartesian equation of p is $x+2y-z=6$





From GC: $\mathbf{r}=\begin{pmatrix}4\\1\\0\end{pmatrix}+\mu\begin{pmatrix}0.5\\0.25\\1\end{pmatrix},\mu\in\mathbb{R}$

(c)

Method 1: B is the intersection of l and p

$$p: \mathbf{r} \bullet \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = 6$$

Since B lies on l, $\overrightarrow{OB} = \begin{pmatrix} 0 \\ 1 \\ -12 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 1 \\ 10 \end{pmatrix}$ for some $\lambda \in \mathbb{R}$

When l and p intersect,

$$\left[\begin{pmatrix} 0 \\ 1 \\ -12 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 1 \\ 10 \end{pmatrix} \right] \bullet \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = 6$$

$$2 + 12 + (4 + 2 - 10)\lambda = 6$$

$$14 - 4\lambda = 6$$

$$\lambda = 2$$

$$\therefore \overrightarrow{OB} = \begin{pmatrix} 0 \\ 1 \\ -12 \end{pmatrix} + 2 \begin{pmatrix} 4 \\ 1 \\ 10 \end{pmatrix} = \begin{pmatrix} 8 \\ 3 \\ 8 \end{pmatrix}$$

Method 2: B is the intersection of l and m

When l and m intersect,

$$\begin{pmatrix} 0 \\ 1 \\ -12 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 1 \\ 10 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}$$

$$4\lambda - 2\mu = 4$$

$$\lambda - \mu = 0$$

Solving, $\lambda = 2, \mu = 2$

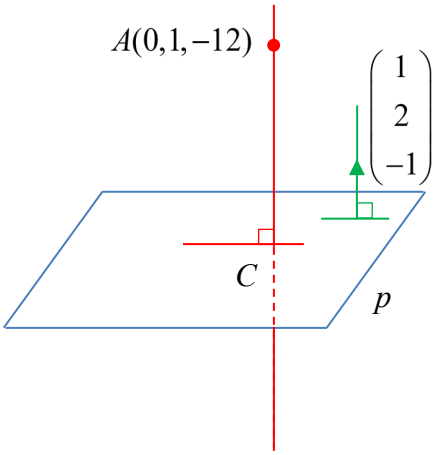
$$\therefore \overrightarrow{OB} = \begin{pmatrix} 0 \\ 1 \\ -12 \end{pmatrix} + 2 \begin{pmatrix} 4 \\ 1 \\ 10 \end{pmatrix} = \begin{pmatrix} 8 \\ 3 \\ 8 \end{pmatrix}$$

(d)

Method 1: C is the foot of perpendicular of A to p

$$p: \mathbf{r} \bullet \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = 6$$

$$l_{AC}: \mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ -12 \end{pmatrix} + s\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad s \in \mathbb{R}$$



When p and l_{AC} intersect,

$$\left[\begin{pmatrix} 0 \\ 1 \\ -12 \end{pmatrix} + s\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}\right] \bullet \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = 6$$

$$2 + 12 + (1 + 2 + 1)s = 6$$

$$14 + 6s = 6$$

$$s = -\frac{4}{3}$$

$$\therefore \overrightarrow{OC} = \begin{pmatrix} 0 \\ 1 \\ -12 \end{pmatrix} - \frac{4}{3}\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} -4 \\ -5 \\ -32 \end{pmatrix}$$

Method 2: C is the foot of perpendicular of A to m

Since C lies on line m,

$$\overrightarrow{OC} = \begin{pmatrix} 4 \\ 1 \\ 0 \end{pmatrix} + \mu\begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}$$

$$\overrightarrow{AC} = \begin{pmatrix} 4 \\ 1 \\ 0 \end{pmatrix} + \mu\begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ -12 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 12 \end{pmatrix} + \mu\begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}$$

Since $\overrightarrow{AC}$ is perpendicular to m,

$$\left[\begin{pmatrix} 4 \\ 0 \\ 12 \end{pmatrix} + \mu\begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}\right] \bullet \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix} = 0$$

$$56 + 21\mu = 0$$

$$\mu = -\frac{8}{3}$$

$$\therefore \overrightarrow{OC} = \begin{pmatrix} 4 \\ 1 \\ 0 \end{pmatrix} - \frac{8}{3}\begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} -4 \\ -5 \\ -32 \end{pmatrix}$$

(e)

Method 1

$\overrightarrow{AB} = 2\begin{pmatrix}4\\1\\10\end{pmatrix}$, from part **(c)**;

$\overrightarrow{AC} = -\frac{4}{3}\begin{pmatrix}1\\2\\-1\end{pmatrix}$, from part **(d)**

Area of $\triangle ABC$

$$= \frac{1}{2}\left|2\begin{pmatrix}4\\1\\10\end{pmatrix} \times \left(-\frac{4}{3}\right)\begin{pmatrix}1\\2\\-1\end{pmatrix}\right|$$

$$= \frac{1}{2}(2)\left(\frac{4}{3}\right)\left|-7\begin{pmatrix}3\\-2\\-1\end{pmatrix}\right| \quad \text{[Note : Cross product done in part \textbf{(a)}]}$$

$$= \frac{28}{3}\sqrt{14} \text{ units}^2$$

Method 2

$\overrightarrow{AC} = -\frac{4}{3}\begin{pmatrix}1\\2\\-1\end{pmatrix}$; $\overrightarrow{BC} = -\frac{14}{3}\begin{pmatrix}2\\1\\4\end{pmatrix}$

Area of $\triangle ABC = \frac{1}{2}\left|\left(-\frac{4}{3}\right)\begin{pmatrix}1\\2\\-1\end{pmatrix}\right|\left|\left(-\frac{14}{3}\right)\begin{pmatrix}2\\1\\4\end{pmatrix}\right|$, since $\angle ACB = 90°$

$$= \frac{1}{2}\left(\frac{4}{3}\right)\left(\frac{14}{3}\right)\sqrt{6}\sqrt{21}$$

$$= \frac{28}{3}\sqrt{14} \text{ units}^2$$

Source of Question: DHS Prelim 9758/2022/01/Q9

3 The plane p_1 contains the point $A(-4,4,0)$ and the line with equation $\frac{x-2}{2}=\frac{y-4}{3}, z=6.$

The plane p_2 has equation $\mathbf{r}=\begin{pmatrix}2\\4\\6\end{pmatrix}+\lambda\begin{pmatrix}1\\0\\-1\end{pmatrix}+\mu\begin{pmatrix}0\\1\\1\end{pmatrix}$, where λ and μ are real constants.

(a) Show that the line of intersection between p_1 and p_2 is parallel to the vector $\begin{pmatrix}5\\6\\1\end{pmatrix}$.

Hence write down the vector equation of the line of intersection between p_1 and p_2. [5]

(b) Determine the acute angle between p_1 and p_2. [2]

(c) Find the position vector of the foot of perpendicular from point A to p_2. [3]

(d) The plane p_3 has equation $ax+3y+2z=b$, where $a,b\in\mathbb{R}$. Find the values of a and b such that all three planes have a common line of intersection. [2]

Solution:

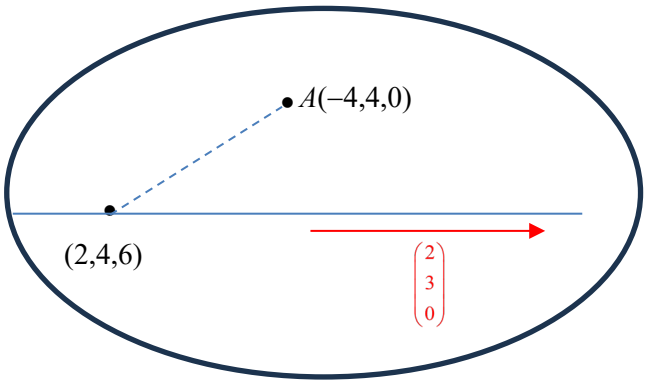
(a)

$\frac{x-2}{2}=\frac{y-4}{3}, z=6.$

To find equationn of the line,

$$\begin{cases}\alpha=\frac{x-2}{2}\Rightarrow x=2+2\alpha\\ \alpha=\frac{y-4}{3}\Rightarrow y=4+3\alpha\\ z=6\end{cases}$$

$$\Rightarrow \mathbf{r}=\begin{pmatrix}2\\4\\6\end{pmatrix}+\alpha\begin{pmatrix}2\\3\\0\end{pmatrix},\alpha\in\mathbb{R}$$



$$\left[\begin{pmatrix}-4\\4\\0\end{pmatrix}-\begin{pmatrix}2\\4\\6\end{pmatrix}\right]\times\begin{pmatrix}2\\3\\0\end{pmatrix}=\begin{pmatrix}-6\\0\\-6\end{pmatrix}\times\begin{pmatrix}2\\3\\0\end{pmatrix}=\begin{pmatrix}18\\-12\\-18\end{pmatrix}=6\begin{pmatrix}3\\-2\\-3\end{pmatrix}.$$

A normal of plane p_1 is $\mathbf{n_1}=\begin{pmatrix}3\\-2\\-3\end{pmatrix}$

$$p_2:\ \mathrm{r}=\begin{pmatrix}2\\4\\6\end{pmatrix}+\lambda\begin{pmatrix}1\\0\\-1\end{pmatrix}+\mu\begin{pmatrix}0\\1\\1\end{pmatrix}$$

Normal of p_2, $\mathbf{n_2}=\begin{pmatrix}1\\0\\-1\end{pmatrix}\times\begin{pmatrix}0\\1\\1\end{pmatrix}=\begin{pmatrix}1\\-1\\1\end{pmatrix}$

A direction vector of line of intersection, $\mathbf{n_2}\times\mathbf{n}_1=\begin{pmatrix}1\\-1\\1\end{pmatrix}\times\begin{pmatrix}3\\-2\\-3\end{pmatrix}=\begin{pmatrix}5\\6\\1\end{pmatrix}$

Hence the line of intersection is parallel to $\begin{pmatrix}5\\6\\1\end{pmatrix}$. (shown)

By observation, $(2,4,6)$ lies on both planes p_1 and p_2.

How?

Look at equation of the given line which lies on p_1:

$$\mathbf{r}=\begin{pmatrix}2\\4\\6\end{pmatrix}+\alpha\begin{pmatrix}2\\3\\0\end{pmatrix},\alpha\in\mathbb{R}$$

Look at equation of the plane p_2 : $\mathrm{r}=\begin{pmatrix}2\\4\\6\end{pmatrix}+\lambda\begin{pmatrix}1\\0\\-1\end{pmatrix}+\mu\begin{pmatrix}0\\1\\1\end{pmatrix}$.

$\therefore$ A vector equation of the line of intersection is $\mathbf{r}=\begin{pmatrix}2\\4\\6\end{pmatrix}+\beta\begin{pmatrix}5\\6\\1\end{pmatrix},\beta\in\mathbb{R}$

(b)

Let acute angle between p_1 and p_2 be θ.

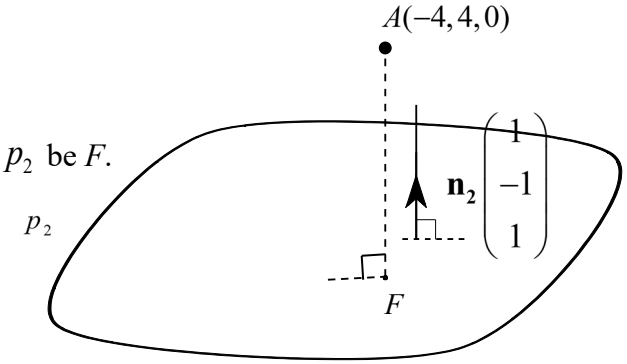
$$\cos\theta = \frac{\left|\begin{pmatrix}1\\-1\\1\end{pmatrix}\cdot\begin{pmatrix}3\\-2\\-3\end{pmatrix}\right|}{\sqrt{3}\sqrt{22}} = \frac{2}{\sqrt{3}\sqrt{22}}$$

$$\theta = 75.7° \text{ (1 dp)}$$

(c)

Method 1 (using scalar product equation of p_2)

$$p_2 : \mathbf{r}\bullet\begin{pmatrix}1\\-1\\1\end{pmatrix} = 4$$



Let the foot of perpendicular from A to p_2 be F.

$$l_{AF} : \mathbf{r} = \begin{pmatrix}-4\\4\\0\end{pmatrix} + \gamma\begin{pmatrix}1\\-1\\1\end{pmatrix},\ \gamma\in\mathbb{R}$$

Since F lies on l_{AF}, $\overrightarrow{OF} = \begin{pmatrix}-4\\4\\0\end{pmatrix} + \gamma\begin{pmatrix}1\\-1\\1\end{pmatrix}$ for some value of γ

Since F also lies on p_2,

$$\left[\begin{pmatrix}-4\\4\\0\end{pmatrix} + \gamma\begin{pmatrix}1\\-1\\1\end{pmatrix}\right]\bullet\begin{pmatrix}1\\-1\\1\end{pmatrix} = 4$$

$$-4-4+(1+1+1)\gamma = 4$$

$$3\gamma = 12$$

$$\gamma = 4$$

$$\therefore \overrightarrow{OF} = \begin{pmatrix}-4\\4\\0\end{pmatrix} + 4\begin{pmatrix}1\\-1\\1\end{pmatrix} = \begin{pmatrix}0\\0\\4\end{pmatrix}$$

Remark :
Take this opportunity to also recall how to find foot of perpendicular from a point to a line.

Method 2 (using vector equation of p_2)

$$l_{AF} : \mathbf{r} = \begin{pmatrix}-4\\4\\0\end{pmatrix} + \gamma\begin{pmatrix}1\\-1\\1\end{pmatrix},\ \gamma\in\mathbb{R};\qquad p_2 :\ \mathbf{r} = \begin{pmatrix}2\\4\\6\end{pmatrix} + \lambda\begin{pmatrix}1\\0\\-1\end{pmatrix} + \mu\begin{pmatrix}0\\1\\1\end{pmatrix}$$

Since F is the point of intersection between l_{AF} and p_2,

$$\begin{pmatrix}-4\\4\\0\end{pmatrix}+\gamma\begin{pmatrix}1\\-1\\1\end{pmatrix}=\begin{pmatrix}2\\4\\6\end{pmatrix}+\lambda\begin{pmatrix}1\\0\\-1\end{pmatrix}+\mu\begin{pmatrix}0\\1\\1\end{pmatrix}$$

$$\lambda\begin{pmatrix}1\\0\\-1\end{pmatrix}+\mu\begin{pmatrix}0\\1\\1\end{pmatrix}+\gamma\begin{pmatrix}-1\\1\\-1\end{pmatrix}=\begin{pmatrix}-4\\4\\0\end{pmatrix}-\begin{pmatrix}2\\4\\6\end{pmatrix}=\begin{pmatrix}-6\\0\\-6\end{pmatrix}$$

$$\lambda\quad-\gamma=-6$$
$$\mu+\gamma=0$$
$$-\lambda+\mu-\gamma=-6$$

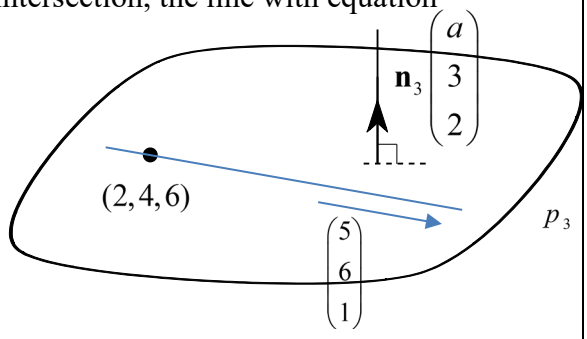
Using the GC to solve the eqns, $\lambda=-2,\ \mu=-4,\ \gamma=4$

Thus, $\overrightarrow{OF}=\begin{pmatrix}-4\\4\\0\end{pmatrix}+4\begin{pmatrix}1\\-1\\1\end{pmatrix}=\begin{pmatrix}0\\0\\4\end{pmatrix}$.

(d)

$p_3:\ ax+3y+2z=b,$ where $a,b\in\mathbb{R}$.

Since all three planes have a common line of intersection, the line with equation $\mathbf{r}=\begin{pmatrix}2\\4\\6\end{pmatrix}+\beta\begin{pmatrix}5\\6\\1\end{pmatrix},\beta\in\mathbb{R}$ must lie on p_3.



Thus the normal of p_3 is perpendicular to the line of intersection:

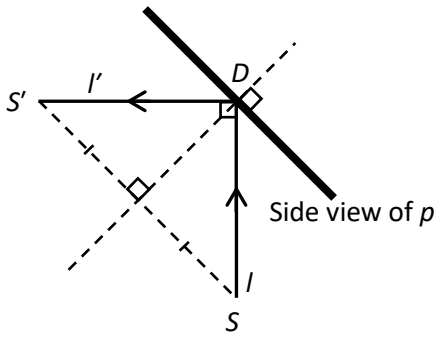
$$\begin{pmatrix}a\\3\\2\end{pmatrix}\cdot\begin{pmatrix}5\\6\\1\end{pmatrix}=0\Rightarrow 5a+18+2=0\Rightarrow a=-4$$

And the point (2, 4, 6) lies on p_3:

$$\begin{pmatrix}a\\3\\2\end{pmatrix}\cdot\begin{pmatrix}2\\4\\6\end{pmatrix}=b\Rightarrow 2a+24=b$$

$\therefore\ a=-4,\ b=16.$

Source of Question: NJC Prelim 9758/2022/01/Q11
4



The diagram above shows a glass prism. The surface of the prism is part of the plane p with equation $by - z = 4$, where b is an integer constant. A ray of light l passes through a point S with coordinates $(5, 1, 3)$, and travels in a direction parallel to the unit vector $\frac{2}{3}\mathbf{i} + c\mathbf{j} + \frac{2}{3}\mathbf{k}$, where c is a negative constant, until it hits the surface of the prism at point D. The light ray is reflected by the surface of the prism. The reflected ray of light, l', passes through the point S', where S' is the image of point S. It is given that the lines l and l' are perpendicular and they lie on the same plane as the normal to p passing through point D. Also, the acute angle between l and the normal to p is equal to the acute angle between l' and the normal to p.

(i) Find the exact value of c and show that $b = 1$. [4]

(ii) Find the coordinates of point D. [3]

(iii) Show that the coordinates of S' are $(m, -1, 1)$ for some integer constant m to be determined and find a vector equation of l' in exact form. [5]

Solution:

(i)

Since $\begin{pmatrix} 2/3 \\ c \\ 2/3 \end{pmatrix}$ is a unit vector, $\left|\begin{pmatrix} 2/3 \\ c \\ 2/3 \end{pmatrix}\right| = 1$.

$$\sqrt{\frac{4}{9} + c^2 + \frac{4}{9}} = 1 \Rightarrow \frac{8}{9} + c^2 = 1$$

$$\Rightarrow c^2 = \frac{1}{9} \Rightarrow c = \pm\frac{1}{3} \Rightarrow c = -\frac{1}{3}$$

$$\left(\text{reject } c = \frac{1}{3} \text{ since } c < 0\right)$$

Thus, l has direction vector $2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

$$\cos 45° = \left| \frac{\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ b \\ -1 \end{pmatrix}}{\sqrt{2^2+(-1)^2+2^2}\sqrt{b^2+1}} \right|$$

$$\frac{1}{\sqrt{2}} = \left| \frac{-b-2}{3\sqrt{b^2+1}} \right|$$

$$\sqrt{\frac{b^2+1}{2}} = \left| \frac{b+2}{3} \right|$$

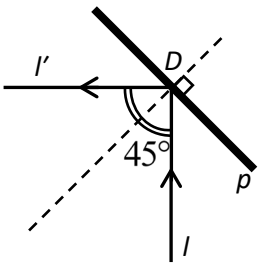
$$\frac{b^2+1}{2} = \left| \frac{b+2}{3} \right|^2 = \frac{b^2+4b+4}{9}$$

$$9b^2+9 = 2b^2+8b+8$$

$$7b^2-8b+1=0$$

$$(b-1)(7b-1)=0$$

$$b=1 \text{ or } b=\frac{1}{7}\left(\text{rej } \because\ b\in\mathbb{Z}\right)$$



RECALL :

1) $\cos\theta = \left| \frac{\mathbf{a}.\mathbf{b}}{|\mathbf{a}||\mathbf{b}|} \right|$, where θ is the acute angle between **a** and **b**.

2) $|x|^2 = x^2$

(ii)

$$l : \mathbf{r} = \begin{pmatrix} 5 \\ 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix},\ \lambda \in \mathbb{R}.$$

Since D lies on l, $\overrightarrow{OD} = \begin{pmatrix} 5+2\lambda \\ 1-\lambda \\ 3+2\lambda \end{pmatrix}$ for some $\lambda \in \mathbb{R}$.

p has equation $y-z=4 \Rightarrow \mathbf{r} \cdot \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = 4$

Since D lies on p, $\begin{pmatrix} 5+2\lambda \\ 1-\lambda \\ 3+2\lambda \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = 4.$

$$\Rightarrow 1-\lambda-3-2\lambda=4$$

$$\Rightarrow 3\lambda=-6$$

$$\Rightarrow \lambda=-2$$

So $\overrightarrow{OD} = \begin{pmatrix} 5+2(-2) \\ 1-(-2) \\ 3+2(-2) \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix}$

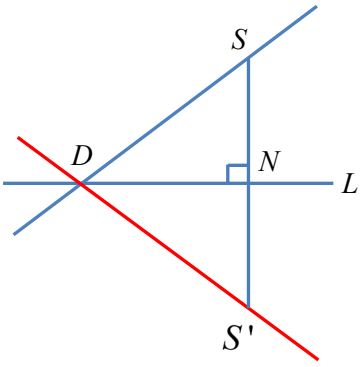
$\therefore$ Coordinates of D are $(1,3,-1)$.

REMARK :
Point D is essentially the point of intersection between the line l and the plane p.

(iii)

Let L be the normal to p passing through D and N be the foot of perpendicular of S onto L.

L has equation $\mathbf{r} = \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$, $\mu \in \mathbb{R}$.



Since N lies on L, $\overrightarrow{ON} = \begin{pmatrix} 1 \\ 3+\mu \\ -1-\mu \end{pmatrix}$ for some $\mu \in \mathbb{R}$.

$$\therefore \overrightarrow{SN} = \begin{pmatrix} 1 \\ 3+\mu \\ -1-\mu \end{pmatrix} - \begin{pmatrix} 5 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} -4 \\ 2+\mu \\ -4-\mu \end{pmatrix}$$

Since $SN \perp \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$, $\overrightarrow{SN} \cdot \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = 0$.

$$\begin{pmatrix} -4 \\ 2+\mu \\ -4-\mu \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = 0$$

$$2+\mu+4+\mu = 0$$

$$2\mu = -6$$

$$\mu = -3$$

$$\therefore \overrightarrow{ON} = \begin{pmatrix} 1 \\ 3+(-3) \\ -1-(-3) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

By Ratio Theorem, $\overrightarrow{ON} = \dfrac{\overrightarrow{OS} + \overrightarrow{OS'}}{2}$.

$$\overrightarrow{OS'} = 2\overrightarrow{ON} - \overrightarrow{OS} = 2\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} - \begin{pmatrix} 5 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} -3 \\ -1 \\ 1 \end{pmatrix} \text{ where } m = -3$$

Thus, S' has coordinates $(-3, -1, 1)$. (shown)

$$\overrightarrow{DS'} = \begin{pmatrix} -3 \\ -1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix} = \begin{pmatrix} -4 \\ -4 \\ 2 \end{pmatrix} = -2\begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$$

$\therefore$ l' has equation $\mathbf{r} = \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix} + \alpha \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$, $\alpha \in \mathbb{R}$.

Source of Question: EJC Prelim 9758/2022/01/Q7

5 The line l_1 has equation $\mathbf{r} = 11\mathbf{i} - 3\mathbf{j} - \mathbf{k} + \lambda(\mathbf{i} - \mathbf{j} + 2\mathbf{k})$, where λ is a parameter. The point A has coordinates $(2, 0, -1)$.

(a) The plane p contains the line l_1 and the point A. Find a cartesian equation of the plane p.

[3]

(b) Find the position vector of the point A', the reflection of the point A in the line l_1. [4]

The line l_2 passes through the point A, and is perpendicular to the line l_1. Planes p and q are perpendicular planes such that p and q meet at line l_2.

(c) Find a vector equation of the line l_2. [2]

(d) The plane $\prod$ is such that $\prod$ contains the point $(11, -3, -1)$ and is parallel to q. Find the perpendicular distance between $\prod$ and q. [2]

Solution:

(a)

Two vectors parallel to p are $\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} - \begin{pmatrix} 11 \\ -3 \\ -1 \end{pmatrix} = 3\begin{pmatrix} -3 \\ 1 \\ 0 \end{pmatrix}$.

$\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \times \begin{pmatrix} -3 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ -6 \\ -2 \end{pmatrix} = -2\begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$.

Thus a vector normal to p is $\begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$.

Since $A(2, 0, -1)$ lies on p, equation of p is $\mathbf{r} \cdot \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} = 1$,

a cartesian equation of p is $x + 3y + z = 1$.

(b)

Let G be the foot of perpendicular from A to l_1.

Since G lies on l_1, $\overrightarrow{OG} = \begin{pmatrix} 11 \\ -3 \\ -1 \end{pmatrix} + \lambda\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ for some λ.

$$\overrightarrow{AG} \cdot \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = 0 \Rightarrow \left[\begin{pmatrix} 11 \\ -3 \\ -1 \end{pmatrix} + \lambda\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}\right] \cdot \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = 0$$

$$\Rightarrow 12 + 6\lambda - 0 = 0$$

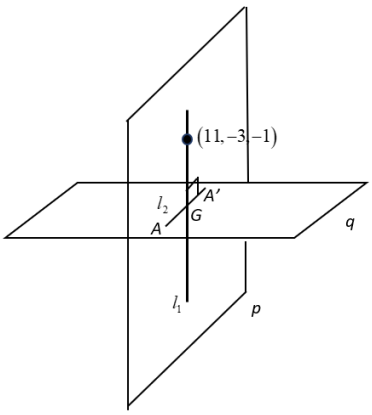
$$\Rightarrow \lambda = -2$$

Hence, $\overrightarrow{OG} = \begin{pmatrix} 9 \\ -1 \\ -5 \end{pmatrix}$.

Using Ratio Theorem, $\overrightarrow{OG} = \dfrac{\overrightarrow{OA} + \overrightarrow{OA'}}{2}$.

$\therefore \quad \overrightarrow{OA'} = 2\overrightarrow{OG} - \overrightarrow{OA} = 2\begin{pmatrix} 9 \\ -1 \\ -5 \end{pmatrix} - \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 16 \\ -2 \\ -9 \end{pmatrix}$.

(c)



A direction vector of l_2 is $\overrightarrow{AG} = \begin{pmatrix} 7 \\ -1 \\ -4 \end{pmatrix}$.

A vector equation of l_2 is

$\mathbf{r} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} + \mu\begin{pmatrix} 7 \\ -1 \\ -4 \end{pmatrix}, \quad \mu \in \mathbb{R}$.

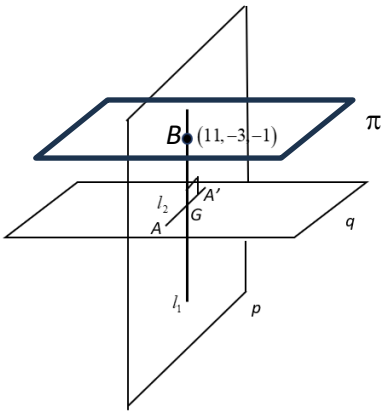
Alternative : l_2 is the line of intersection between p and q

$p : x + 3y + z = 1$ --- (1)

$q : \mathbf{r} \cdot \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = 0 \Rightarrow x - y + 2z = 0$ --- (2)

Using GC to solve (1) and (2), an equation of l_2 is $\mathbf{r} = \begin{pmatrix} \frac{1}{4} \\ \frac{1}{4} \\ 0 \end{pmatrix} + \mu\begin{pmatrix} -\frac{7}{4} \\ \frac{1}{4} \\ 1 \end{pmatrix}, \quad \mu \in \mathbb{R}$.

(d)



Since $\overrightarrow{GB}=\begin{pmatrix}11\\-3\\-1\end{pmatrix}-\begin{pmatrix}9\\-1\\-5\end{pmatrix}=\begin{pmatrix}2\\-2\\4\end{pmatrix}=2\begin{pmatrix}1\\-1\\2\end{pmatrix}$, $\overrightarrow{GB}$ is parallel to l_1. As G is on l_1, $B(11,-3,-1)$ is also on l_1 and l_1 is perpendicular to both q and Π.

Perpendicular distance between Π and $q = \left|\overrightarrow{GB}\right| = \left|\begin{pmatrix}11\\-3\\-1\end{pmatrix}-\begin{pmatrix}9\\-1\\-5\end{pmatrix}\right| = \left|\begin{pmatrix}2\\-2\\4\end{pmatrix}\right| = 2\sqrt{6}$ units

Alternative 1 : Find projection length of $\overrightarrow{AB}$ on normal)

Perpendicular distance $= \dfrac{\left|\overrightarrow{AB}\cdot\begin{pmatrix}1\\-1\\2\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\-1\\2\end{pmatrix}\right|} = \dfrac{\left|\begin{pmatrix}9\\-3\\0\end{pmatrix}\cdot\begin{pmatrix}1\\-1\\2\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\-1\\2\end{pmatrix}\right|} = \dfrac{12}{\sqrt{6}}$ units

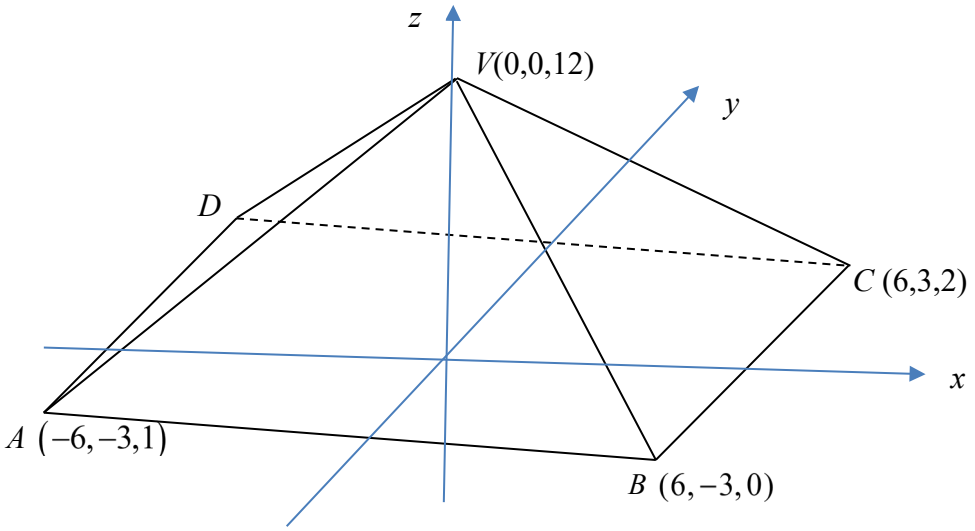
Alternative 2 : Use distance between parallel planes formula

Plane Π: $\mathbf{r}\cdot\begin{pmatrix}1\\-1\\2\end{pmatrix} = \begin{pmatrix}11\\-3\\-1\end{pmatrix}\cdot\begin{pmatrix}1\\-1\\2\end{pmatrix} = 12$ Plane q: $\mathbf{r}\cdot\begin{pmatrix}1\\-1\\2\end{pmatrix} = 0$

Perpendicular distance $= \dfrac{|d_1 - d_2|}{|\mathbf{n}|} = \dfrac{|12-0|}{\left|\begin{pmatrix}1\\-1\\2\end{pmatrix}\right|} = \dfrac{12}{\sqrt{6}}$ unit

Source of Question: SAJC CT2 9758/2023/02/Q2

6



An oblique pyramid has a plane base $ABCD$ in the shape of a parallelogram. The coordinates of A, B and C are $(-6,-3,1)$, $(6,-3,0)$ and $(6,3,2)$ respectively. The apex of the pyramid is at $V(0,0,12)$.

(i) Find the coordinates of D. [1]

(ii) Find the cartesian equation of face BCV. [3]

(iii) Find the angle between face BCV and the base of the pyramid. [3]

(iv) Find the shortest distance from the midpoint of edge AD to plane BCV. [3]

Solution:

(i)

$$\overrightarrow{AD} = \overrightarrow{BC}$$

$$\overrightarrow{OD} - \overrightarrow{OA} = \overrightarrow{OC} - \overrightarrow{OB}$$

$$\overrightarrow{OD} = \overrightarrow{OC} - \overrightarrow{OB} + \overrightarrow{OA}$$

$$\overrightarrow{OD} = \begin{pmatrix} 6 \\ 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 6 \\ -3 \\ 0 \end{pmatrix} + \begin{pmatrix} -6 \\ -3 \\ 1 \end{pmatrix} = \begin{pmatrix} -6 \\ 3 \\ 3 \end{pmatrix}$$

$\therefore D(-6,3,3)$

(ii)

$$\overrightarrow{VC} = \begin{pmatrix} 6 \\ 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 12 \end{pmatrix} = \begin{pmatrix} 6 \\ 3 \\ -10 \end{pmatrix}$$

$$\overrightarrow{VB} = \begin{pmatrix} 6 \\ -3 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 12 \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \\ -12 \end{pmatrix} = 3\begin{pmatrix} 2 \\ -1 \\ -4 \end{pmatrix}$$

	$\overrightarrow{VC}\times\overrightarrow{VB}\,//\begin{pmatrix}6\\3\\-10\end{pmatrix}\times\begin{pmatrix}2\\-1\\-4\end{pmatrix}=\begin{pmatrix}-22\\4\\-12\end{pmatrix}=2\begin{pmatrix}-11\\2\\-6\end{pmatrix}$ Normal vector of plane BCV is $\begin{pmatrix}-11\\2\\-6\end{pmatrix}$ Therefore the equation of plane BCV is $\underset{\sim}{r}\bullet\begin{pmatrix}-11\\2\\-6\end{pmatrix}=\begin{pmatrix}0\\0\\12\end{pmatrix}\bullet\begin{pmatrix}-11\\2\\-6\end{pmatrix}=-72$ $\therefore$ Cartesian equation of plane BCV is $-11x+2y-6z=-72$
(iii)	$\overrightarrow{BC}=\begin{pmatrix}6\\3\\2\end{pmatrix}-\begin{pmatrix}6\\-3\\0\end{pmatrix}=\begin{pmatrix}0\\6\\2\end{pmatrix}=2\begin{pmatrix}0\\3\\1\end{pmatrix}$ $\overrightarrow{AB}=\begin{pmatrix}6\\-3\\0\end{pmatrix}-\begin{pmatrix}-6\\-3\\1\end{pmatrix}=\begin{pmatrix}12\\0\\-1\end{pmatrix}$ $\overrightarrow{BC}\times\overrightarrow{AB}\,//\begin{pmatrix}0\\3\\1\end{pmatrix}\times\begin{pmatrix}12\\0\\-1\end{pmatrix}=\begin{pmatrix}-3\\12\\-36\end{pmatrix}=3\begin{pmatrix}-1\\4\\-12\end{pmatrix}$ Normal vector of base of the pyramid is $\begin{pmatrix}-1\\4\\-12\end{pmatrix}$ Let the angle between the planes be θ $\cos\theta=\dfrac{\left\lvert\begin{pmatrix}-1\\4\\-12\end{pmatrix}\bullet\begin{pmatrix}-11\\2\\-6\end{pmatrix}\right\rvert}{\sqrt{161}\sqrt{161}}=\dfrac{91}{161}$ $\theta=55.6^0$ (1dp)
(iv)	Let $\overrightarrow{OM}$ be the midpoint of AD $\overrightarrow{OM}=\dfrac{\begin{pmatrix}-6\\-3\\1\end{pmatrix}+\begin{pmatrix}-6\\3\\3\end{pmatrix}}{2}=\begin{pmatrix}-6\\0\\2\end{pmatrix}$

$$\overrightarrow{MV} = \begin{pmatrix} 0 \\ 0 \\ 12 \end{pmatrix} - \begin{pmatrix} -6 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 0 \\ 10 \end{pmatrix}$$

Distance from M to the plane BCV

$= \text{length of projection of } \overrightarrow{MV} \text{ on } \begin{pmatrix} -11 \\ 2 \\ -6 \end{pmatrix}$

$$= \frac{\left| \overrightarrow{MV} \bullet \begin{pmatrix} -11 \\ 2 \\ -6 \end{pmatrix} \right|}{\sqrt{11^2 + 2^2 + 6^2}} = \frac{\left| \begin{pmatrix} 6 \\ 0 \\ 10 \end{pmatrix} \bullet \begin{pmatrix} -11 \\ 2 \\ -6 \end{pmatrix} \right|}{\sqrt{161}} = \frac{126}{\sqrt{161}} = 9.93 \text{ (3sf)}$$